科學HW6.

$u = t^z$, $du = z t^{z-1} dt$

※11.7 $\Gamma(z+1) = \int_0^\infty e^{-t} t^{z+1-1} dt = \int_0^\infty e^{-t} t^{z} dt$. Using Integration by part, $dv = e^{-t} dt$, $v = -e^{-t}$

$\Rightarrow \Gamma(z+1) = -t^z e^{-t}\Big|_0^\infty + z\int_0^\infty e^{-t} t^{z-1} dt = 0 + z\int_0^\infty e^{-t} t^{z-1} dt = z\Gamma(z)$ ※

※11.9 The Local truncation error: $T_n(h) = \dfrac{y(x_{n+1}) - y(x_n)}{h} - \left[(1-\frac{\alpha}{2}) f(x_n, y(x_n)) + \frac{\alpha}{2} f(x_{n+1}, y(x_{n+1}))\right]$

By Taylor expansion, $y(x_{n+1}) - y(x_n) = h y'(x_n) + h^2 \cdot \frac{1}{2} y''(x_n) + \frac{h^3}{6} y'''(x_n) + O(h^4)$

$$\Rightarrow \frac{y(x_{n+1}) - y(x_n)}{h} = y'(x_n) + \frac{h}{2} f''(x_n) + \frac{h^2}{6} y'''(x_n) + O(h^3)$$

and $f(x_{n+1}, y(x_{n+1})) = y'(x_{n+1}) = y'(x_n) + h y''(x_n) + \frac{h^2}{2} y'''(x_n) + O(h^3)$

Thus $(1-\frac{\alpha}{2}) f(x_n, y(x_n)) + \frac{\alpha}{2} f(x_{n+1}, y(x_{n+1})) = (1-\frac{\alpha}{2}) y'(x_n) + \frac{\alpha}{2}\left[y'(x_n) + h y''(x_n) + \frac{h^2}{2} y'''(x_n)\right] + O(h^3)$

$\therefore (1-\frac{\alpha}{2}) f(x_n, y(x_n)) + \frac{\alpha}{2} f(x_{n+1}, y(x_{n+1})) = y'(x_n) + \frac{\alpha}{2} h y''(x_n) + \frac{\alpha}{2} h^2 y'''(x_n) + O(h^3)$

Hence $T_n(h) = \left[y'(x_n) + \frac{h}{2} y''(x_n) + \frac{h^2}{6} y'''(x_n)\right] - \left[y'(x_n) + \frac{\alpha}{2} h y''(x_n) + \frac{\alpha}{2} h^2 y'''(x_n) + O(h^3)\right]$

$= \frac{h}{2}(1-\alpha) y''(x_n) + h^2 (\frac{1}{6} - \frac{\alpha}{4}) y'''(x_n) + O(h^3)$

If $T_n(h) \to 0$ as $h \to 0$, then: ① $\alpha = 1$, $T_n(h) = O(h^2) \Rightarrow$ order 2

② $\alpha \neq 1$, $T_n(h) = O(h) \Rightarrow$ consistent.

For $\alpha = 1$, $u_{n+1} = u_n + \frac{h}{2}\left[f(x_n, u_n) + f(x_{n+1}, u_{n+1})\right]$, and to solve the I.V.P: $\begin{cases} y'(x) = -10 y(x) \\ y(0) = 1 \end{cases}$, $x > 0$

$\Rightarrow u_{n+1} = u_n + \frac{h}{2}\left[-10 u_n - 10 u_{n+1}\right] \Rightarrow u_{n+1} + 5h u_{n+1} = u_n - 5h u_n$

$\Rightarrow (1+5h) u_{n+1} = (1-5h) u_n \Rightarrow u_{n+1} = \dfrac{1-5h}{1+5h} u_n$.

Consider $R(z) = \dfrac{1+\frac{z}{2}}{1-\frac{z}{2}}$, since $y' = \lambda y$ give us $u_{n+1} = \dfrac{1+\frac{h\lambda}{2}}{1-\frac{h\lambda}{2}} u_n \Rightarrow$ We get $z = -10h$ from above.

Hence $R(z) = \dfrac{1-5h}{1+5h}$, for $h \geq 0$, $1-5h \leq 1+5h \Rightarrow |1-5h| \leq |1+5h|$ for $h > 0$.

① If $0 < h < 1/5$, $1-5h > 0 \Rightarrow 0 < R(z) < 1 \Rightarrow |R(z)| < 1$

② If $h = \frac{1}{5}$, $R(z) = 0 \Rightarrow |R(z)| = 0 \leq 1$

③ If $h > \frac{1}{5}$, $1-5h < 0$, then $|R(z)| = \dfrac{5h-1}{5h+1} < 1$ for all $h > 0$

$\therefore$ For $h > 0$, $|R(z)| < 1$, this method is absolutely stable for the I.V.P.

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